Do Artificial Stakes Improve LLM Risk Reasoning?

Shrey Katyal, Dev Sinha, Shreekar Earanti,
Aditya Dubey, Atiksh Shah, Sukrut Chikodikar, Taran Savoor

May 8, 2026

1 Introduction

Large language models (LLMs) are increasingly being used in settings that require careful reasoning, including financial analysis, risk assessment, medical analysis, and other high-stakes domains. In these environments, the quality of a model's decision matters because it must not only select the correct answer, but also reason about probabilities, tradeoffs, and potential downside risk. However, current LLMs do not face meaningful consequences when they make poor decisions. Even when they are asked to reason through a problem, their outputs are not connected to any real cost, penalty, or resource constraint. This creates a gap between how LLMs are evaluated and how human decision-makers operate, since humans often make risk-sensitive decisions under limited budgets, reputational stakes, or material consequences.

This project investigates whether introducing artificial stakes can improve LLM risk reasoning. In our setting, artificial stakes are created through a persistent virtual budget that is shown to the model and updated across a sequence of decisions. Instead of treating each question as an isolated prompt, the model is placed in a decision-making environment in which poor choices reduce its available budget and strong decisions preserve or improve it. The goal is to test whether a persistent resource frame alters how an LLM approaches risk-centered reasoning problems.

Our motivation comes from several limitations in current LLM reasoning behavior. Prior work suggests that LLMs can be overconfident and can produce reasoning traces that sound convincing without necessarily reflecting better underlying decisions. Lanham et al. [1] support this concern by showing that chain-of-thought explanations can be unfaithful to a model's actual decision process, making it important to evaluate whether artificial stakes improve decisions themselves rather than merely producing more plausible explanations.

We hypothesize that models exposed to persistent resource framing will make better risk-sensitive decisions than models given equivalent tasks and feedback without any stakes. We expect the persistent-stakes condition to improve decision accuracy, probability calibration, and sequential decision quality relative to baseline conditions. Importantly, our hypothesis is not merely that models will produce longer or more cautious explanations. Rather, we aim to test whether artificial stakes cause models to make substantively different and better decisions when evaluated against

ground-truth data. By comparing bare baselines, surface-level stakes, persistent stakes, and chain-of-thought controls, we can isolate whether any improvement comes from the persistent budget mechanism itself rather than from additional reasoning language or prompt complexity.

Overall, this project explores whether artificial consequences make LLMs reason more carefully about risk. If persistent virtual stakes do improve model performance, this would suggest a lightweight prompting and evaluation strategy for improving LLM behavior in domains where uncertainty and downside risk matter. If they do not, the results would still be informative, because they would suggest that models may imitate risk-aware language without meaningfully improving their decisions. More broadly, the experiment helps clarify whether consequence framing can serve as a useful tool for evaluating and improving LLM reasoning in realistic, high-stakes decision environments.

1.1 Hypothesis

We hypothesize that persistent resource framing—where a model observes a virtual budget that updates based on the quality of its decisions across a sequence of risk-sensitive problems will produce better decisions than conditions that lack persistent stakes. We predict three effects relative to the bare baseline, surface-stakes, and chain-of-thought control conditions:

First, models in the persistent-stakes condition will achieve higher decision accuracy against ground truth on both single-shot risk-reasoning items (expected value, conditional probability, and Bayesian update problems) and on real-world benchmarks in finance, science, and medicine.

Second, persistent stakes will improve probability calibration, resulting in lower expected calibration error and Brier scores as models become less overconfident when poor decisions deplete their budget.

Third, in the sequential allocation environment, persistent stakes will increase cumulative reward, reduce regret, and lower the conditional probability of compounding mistakes, indicating that the budget signal shapes a decision policy rather than isolated responses.

We hypothesize that any observed gains will reflect substantively different decisions rather than stylistic shifts in reasoning language. We therefore predict that the persistent-stakes condition will outperform the chain-of-thought control on accuracy and calibration metrics, even though both conditions elicit longer and more deliberative reasoning traces. If instead the persistent-stakes and chain-of-thought conditions perform comparably, this would suggest that consequence framing primarily causes more cautious sounding language without altering any underlying decision quality.

2 Related Work

One of the main reasons we chose this research topic was because we noticed that current language models don't really have an incentive to make smarter, more measured decisions. We hypothesized that creating a constrained budget environment allowed models to gauge their own uncertainty, and by punishing bad decisions, we could "force" them to make better ones. In our quest to support

our ideas, we found many interesting papers that allowed us to shape our understanding and give our hypothesis some traction.

Logically, we first considered existing reward models whose alignment training was targeted at pushing models to reason better. However, techniques like RLHF actually made overconfidence worse, since reward models tend to favor confident-sounding responses regardless of quality (Leng et al.) [3]. This works against what we wanted entirely. Reinforcement learning over reasoning steps doesn't fix it either, since RL-trained models are prone to motivated reasoning, generating plausible-sounding justifications for violating their instructions rather than actually reasoning through them (Howe & Carroll) [4]. Both approaches try to shape behavior through training. Ours is different: we don't touch the model at all and instead let decision correctness do the shaping at the prompt level.

Once we landed on a prompting-based approach, we had to think carefully about how to actually measure whether it worked. This turned out to be harder than expected because chain-of-thought traces often don't reflect how a model actually arrived at its answer (Lanham et al.) [1]. Lanham et al. show that models can produce coherent reasoning traces that have no real bearing on the answer they actually give. So rather than trusting the reasoning trace, we anchor evaluation entirely to whether the decision itself matches ground truth. Metrics like risk factors surfaced, alternatives considered, and logical error rate still mattered to us, but only as secondary measures to help explain how behavior changes.

We wanted to test our theory on settings that were risk-sensitive and required a model to make deliberate, calculated decisions. We needed domains with verifiable ground-truth answers where the right choice actually requires reasoning about probabilities and tradeoffs. Finance and medicine naturally fit here. As a bonus, LLMs already struggle in these settings, with most agents failing to beat a simple buy-and-hold baseline in realistic trading environments (Li et al.) [5]. InvestorBench confirmed this pattern across thirteen models and multiple asset classes. That benchmark targets risk-sensitive decision quality similar to our setup, but it only characterizes how models fail without testing any intervention to fix it. That told us this was a hard enough target worth testing against, and that there was still room for something like our approach.

Prior work shows that LLMs exhibit systematic behavioral biases, including loss aversion, anchoring, and framing effects, that lead to inconsistent and often irrational economic decisions (Ross et al.) [6]. These tendencies are not reliably corrected by surface-level prompting. Ross et al. document the problem well, but don't propose a fix. That sent us looking for prior work that had actually tried to correct for it. The closest thing we found was BudgetThinker, which showed that conditioning models on token-budget signals can improve accuracy by 4.9% (Wen et al.) [7]. Both approaches use an external resource signal to influence model behavior at inference time, so the similarity is real. But BudgetThinker is about controlling reasoning length, not tying a resource state to whether decisions are actually correct, and as far as we can tell, no prior work has tested that before.

3 Methodological Development

To test whether artificial stakes improve LLM risk reasoning, we developed an experimental pipeline with two complementary task types: controlled decision problems and a sequential budget allocation environment. The same evaluation items are shown under four prompting conditions and scored against fixed ground-truth answers using accuracy, calibration measures, and optional automated judging of reasoning quality.

3.1 Experimental Conditions

We compare four conditions: a plain baseline with no stakes or extra reasoning instructions, a surface-stakes condition where a budget or token cost is mentioned but does not update, a persistent-stakes condition where the model's choices affect a virtual budget that carries across turns, and a chain-of-thought control that encourages structured reasoning without stakes language. This setup helps separate the effect of persistent stakes from simply asking the model to reason more carefully.

3.2 Tasks and Data

The synthetic task generator produces several families of risk-reasoning problems, including risky-versus-safe expected value comparisons, three-option allocation problems, conditional probability questions, base-rate problems, and Bayesian update questions. Each item includes a prompt, answer choices, the correct answer, an explanation, and metadata such as expected values or posterior probabilities. This allows decision accuracy to be scored exactly and reduces ambiguity in the evaluation.

We also include realistic benchmark-style tasks in finance, science, and medical multiple-choice settings. In addition, we evaluate a sequential allocation environment where each episode begins with a fixed virtual budget. At each step, the model chooses among conservative, medium, aggressive, or skip actions. Non-skip actions sample an outcome from a known reward distribution and update the budget, which is then shown in the next prompt. The model sees a visible resource state that persists across decisions. This directly operationalizes the main hypothesis: if artificial stakes matter, then persistent resource framing should change the model's decisions across a sequence, not only its one-shot explanation style.

3.3 Models and Runs

We evaluate open-weight subject models and compare base versus instruct variants where possible, since post-training may change how models respond to consequence framing. Each configuration is run with multiple random seeds and, where useful, several independent completions per item so we can compare single-answer behavior with aggregated responses such as majority vote.

3.4 Outcomes and Analysis

Decision quality is measured by agreement with the reference answer, using exact choice match or numeric checks within tolerance. Calibration is evaluated by comparing stated confidence to empirical correctness using measures such as Brier score, expected calibration error, and reliability-style summaries.

For reasoning quality, a stronger frontier model can act as an LLM judge on the full output, scoring coherence and whether the response considers major risks and alternatives. For sequential episodes, we also log regret, cumulative reward, budget efficiency, recovery rate, and mistake propagation to examine whether early decisions affect later performance.

3.5 Implementation and Scope

The environment is configuration-driven, with conditions, datasets, models, seeds, decoding settings, and metrics logged so runs can be reproduced and compared fairly. We also tested the budget tracker and sequential environment to confirm that budgets initialize and reset correctly, rewards update as expected, invalid actions fail, skip actions leave the budget unchanged, episodes terminate properly, and seeded runs are reproducible.

Because the stakes are simulated rather than real incentives, the conclusions are limited to model behavior under controlled prompting conditions. Judge-based scores are treated as supporting evidence rather than definitive proof of improved reasoning.

4 Results

4.1 Accuracy on Static Datasets

Table 1 and Table 2 report accuracy by condition and dataset for the base and instruct variants of Llama-3.1-8B, respectively. The results reveal a sharp divergence between the two model types that partially rejects our original hypothesis while supporting a more conditional version of it.

Table 1: Accuracy by condition — Llama-3.1-8B Base (25 items per dataset, seed 42)

Dataset	Baseline	Surface Stakes	Persistent Stakes	CoT Only
Synthetic EV	0.24	0.08	0.00*	0.36
ConvFinQA	0.04	0.16	0.14*	0.08
ARC-Challenge	0.28	0.24	0.08*	0.20
MedQA	0.08	0.08	0.08*	0.08

*Budget depleted to $0 before all items were completed (10–14 of 25 items answered).

For the base model, persistent stakes caused budget collapse on all four datasets: the model answered questions incorrectly at a high enough rate that the virtual budget depleted to 0, terminating

each run early. This is a direct rejection of the accuracy sub-hypothesis for base models. Surface stakes produced no consistent improvement, and CoT only modestly outperformed baseline on Synthetic EV.

Table 2: Accuracy by condition — Llama-3.1-8B Instruct (25 items per dataset, seed 42)

Dataset	Baseline	Surface Stakes	Persistent Stakes	CoT Only
Synthetic EV	0.64	0.64	**0.76**	0.64
ConvFinQA	0.24	0.32	**0.48**	0.28
ARC-Challenge	0.76	0.68	**0.80**	0.76
MedQA	**0.56**	0.56	0.48	0.48

For the instruct model, the picture is substantially different. Persistent stakes achieved the highest accuracy on three of four datasets, with gains of +12 percentage points on Synthetic EV, +24 pp on ConvFinQA, and +4 pp on ARC-Challenge relative to baseline. MedQA was the exception, where persistent stakes slightly underperformed baseline (0.48 vs. 0.56). Surface stakes again produced no reliable improvement, and CoT only matched but did not exceed baseline accuracy despite producing higher-quality reasoning traces (see Section 4.4).

These results suggest that persistent stakes function as a lightweight decision-forcing mechanism, but require a base level of instruction-following capability to function. The instruct model uses the budget signal to regulate its behavior; the base model lacks the scaffolding to interpret or act on it productively.

4.2 Calibration

Table 3 reports Expected Calibration Error (ECE) and Brier scores across conditions for both models. Our second sub-hypothesis predicted that persistent stakes would reduce overconfidence and improve calibration.

Table 3: Calibration metrics by condition (Synthetic EV dataset)

Condition	Base ECE	Base Brier	Instruct ECE	Instruct Brier
Baseline	0.655	0.620	0.270	0.271
CoT Only	0.420	0.392	0.282	0.293
Surface Stakes	0.678	0.651	0.275	0.291
Persistent Stakes	0.751	0.680	0.285	0.286

The calibration hypothesis is rejected for both models, though for different reasons. The base model is severely miscalibrated across all conditions (ECE 0.42–0.75), and persistent stakes make calibration worse rather than better (ECE 0.751 vs. 0.655 at baseline). The instruct model is already well-calibrated across all conditions (ECE $\approx$ 0.27–0.28), and stakes framing does not meaningfully change this. Notably, CoT only produces the largest calibration improvement for the base model (ECE 0.420), suggesting that structured reasoning prompts are more effective than consequence framing for reducing overconfidence. This is consistent with the interpretation that instruction-tuning handles most of the calibration benefit for capable models, leaving little room for stakes to add further improvement.

4.3 Sequential Budget Environment

Table 4 reports cumulative regret, mean episode reward, and budget efficiency for the sequential allocation environment, which was designed to most directly operationalize the persistent-stakes hypothesis.

Table 4: Sequential budget environment metrics (3 episodes × 20 steps, seed 42)

Metric	Base Baseline	Base Persistent	Instruct Baseline	Instruct Persistent
Cumulative Regret (final)	4.00	**0.61**	−0.84	**−1.83**
Mean Episode Reward	0.00	3.39	4.84	5.83
Budget Efficiency	1.00	1.43	1.60	1.60

The setting where events unfold step by step most clearly backs the idea that ongoing stakes remain relevant. Here, Starting from the base model, persistent stakes bring down final cumulative regret significantly. From 4.00 it drops to just 0.61. That gap shows a meaningful shift. Notably, ongoing commitment alters outcomes over time. The improvement appears far from random. Instead, it follows consistent engagement. Results dip sharply when persistence is applied. This pattern holds across repeated trials. Though subtle at first, effects grow clear later on. Ultimately, sustained involvement reshapes performance Yet improvement appeared, despite collapse within static data settings. Training followed instruction-focused methods Across trials, persistent stakes show lower cumulative regret than baseline levels, reaching 1.83 compared to 0.84 on average Despite fluctuations, performance exceeds predictions over time. Average returns rise steadily through each trial Under ongoing pressure, performance shifts stand out: 3.39 compared to 0.00 in the base model; meanwhile, the instruct version moves from 4.84 up to 5.83.

Looking at these outcomes, it appears the ongoing budget cue shows strongest impact across multiple exchanges Outcomes become clear right after a choice unfolds, shaping what comes next. Each step forward reveals the impact of prior moves. What happens follows directly from earlier actions taken. Clarity emerges through immediate feedback in these settings. The result of one move shows up in the following moment. Following decisions depend on what just occurred. Feedback appears without delay, tied closely to choices made As steps progress, resources build up gradually; through this, the model sees how its financial path unfolds, leading to improved decisions Later on, decisions become more cautious yet carefully adjusted step by step. That aspect helps restore some value to long-standing patterns One idea suggests the system functions - yet unfolds more clearly across steps instead of one single trial tings

4.4 Reasoning Quality

Table 5 reports coherence and reasoning coverage scores assigned by a Claude Sonnet judge, which evaluated whether responses considered relevant risk factors and alternative choices.

CoT only consistently produces the highest coherence and reasoning coverage scores on both models. Stakes conditions, whether surface or persistent, do not improve reasoning quality beyond baseline levels. For the instruct model, persistent stakes actually produce the lowest reasoning coverage score (0.693) despite achieving the highest accuracy on most datasets.

Table 5: Judge scores (coherence and reasoning coverage, scale 0–1)

Condition	Base Coherence	Base Coverage	Instruct Coherence	Instruct Coverage
Baseline	0.134	0.080	0.758	0.716
CoT Only	**0.213**	**0.195**	**0.795**	**0.749**
Surface Stakes	0.138	0.101	0.785	0.722
Persistent Stakes	0.140	0.080	0.779	0.693

This split between how well a model reasons and how good its choices are matters. Because high-stakes conditions lead to better outcomes without clearer thinking, possibly by increasing reluctance toward uncertain responses, instead of deepening thought. So while consequences do influence results, they skip enhancing logic entirely. Their role feels closer to a filter on behavior than support for reflection.

5 Conclusion & Future Work

Our results do not support the hypothesis that artificial stakes reliably improve LLM risk reasoning. If the hypothesis were true in a broad sense, we would expect surface or persistent stakes to consistently improve accuracy, calibration, or reasoning quality across datasets and model types. Instead, the effect of stakes was inconsistent. In the base model, stakes often hurt performance, while in the instruct model they sometimes helped but the improvement was irregular across domains. This suggests that adding consequence-like framing alone is not reliable enough to make models reason more carefully, and may sometimes distract the model and make it overly cautious or unstable in terms of its behavior.

A stronger result that we can infer is that reasoning scaffolding matters more than artificial stakes. CoT-only often improved reasoning behavior as evaluated by the Claude Sonnet judge, such as considering alternatives and risk factors, even when it did not always produce the highest accuracy. This weakens the original "stakes improve reasoning" hypothesis. Stakes may only help when paired with structured reasoning procedures or when the task environment makes consequences concrete, as in the sequential-budget setting.

Future work would involve refining the hypothesis rather than discarding it entirely. A good place to start is to see whether artificial stakes improve reasoning only in sequential environments, only for instruction-tuned models, or only when paired with explicit decision-making structure such as expected-value calculations, budget tracking, or recovery planning. Larger runs across more models and datasets would also help determine whether the mixed effects observed here are noise or evidence that stakes are fundamentally less reliable than direct reasoning prompts.

References

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